

AI-powered code optimisation in R

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Talk overview

- Using AI to help with code writing (Sep 2025)
- Using AI to help with code writing (Sep 2026)
- A sneak peek into VS code
- What have I learnt?



Context

- 15+ years of using R (pre-tidyverse; pre-Stack Overflow)
- Previously in health economics / medical statistics
- Currently in hypothetical higher education institution
- Large datasets, linked patient-/student-level data
- Similar problems: complex/missing data, reproducibility, code scalability...
- ChatGPT Edu, GitHub Copilot Pro, Gemini
- Query punctuation and spelling preserved



Source: https://oxfordvisit.com/articles/harry-potter-filming-locations-great-hall-christ-church/#google_vignette

Using AI to help with code writing (Sep 2025)

The code that you always forget

how to apply the same operation to several columns in R tidyverse

give me an example of names_pattern argument in pivot_wider in tidyverse

How to construct a regular expression in R containing letter b and not containing letter a

Give me an example of applying across() to a group of lab markers

- Short queries;
- excellent concise examples, detailed explanations;
- more useful than search engines search due to interactivity;
- helpful suggestions.

The code that you want to adapt

Convert this flextable code <...> into kable syntax

Using tidyverse, I want to parameterise this code: <...> I have several columns I want to apply this case_when to

I now want to keep column names and supply variables as a parameter vector

now I want to transpose, have gender on top, variable at the bottom. result is still a data.frame

- (most) useful when done iteratively and in small incremental steps

New code/functionality

explain the margin argument in the plot layout command? I am combining two ggplot objects with plotly

kable. I need to left-align heading produced by pack-rows. can I do it with css?

I am writing a quarto revealjs presentation. I need to provide slides in a 16:9 ratio. how can I do this

- Reasonably good experience, BUT:
- good in producing code that looks plausible; hallucinations with niche requests;
- code might still be executable but still not do what was asked;
- fine-tuning the query and/or running through several LLMs might help;
- **code and results always need to be checked.**

Plausible code vs working code

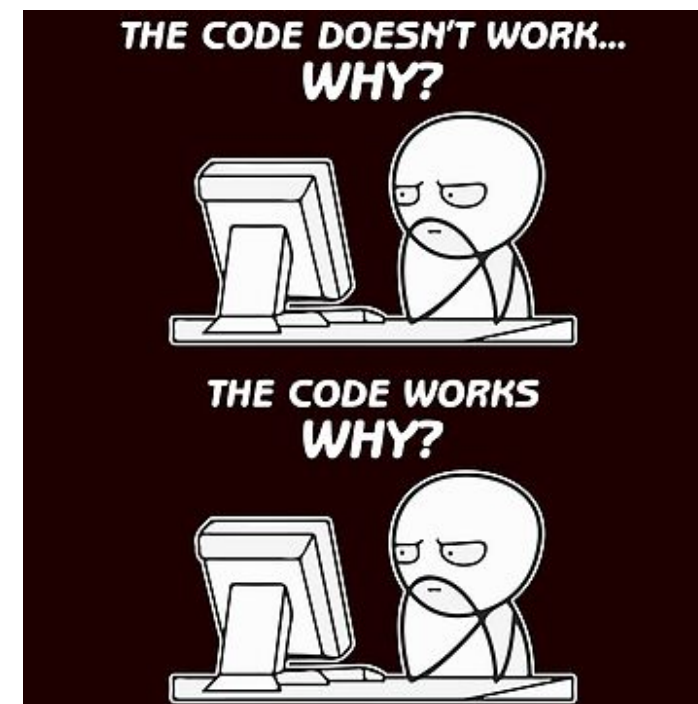
quarto html; how can I generate list of tables & figures?

This is my YAML <...>. Add crossref: true there

error: (line 4, columns 11–15) Field “crossref” has value true, which must instead be ‘false’

but then nothing is produced

i added this but no list was produced



Summary (Sep 2025)

AI is an answer to this!

NEVER HAVE I FELT SO
CLOSE TO ANOTHER SOUL
AND YET SO HELPLESSLY ALONE
AS WHEN I GOOGLE AN ERROR
AND THERE'S ONE RESULT
A THREAD BY SOMEONE
WITH THE SAME PROBLEM
AND NO ANSWER
LAST POSTED TO IN 2003



Source: https://imgs.xkcd.com/comics/wisdom_of_the_ancients.png

Using AI to help with code writing (Sep 2026)

Same old

- The code that you always forget
- The code that you want to adapt
- New code/functionality

- My baseline knowledge has changed
- My confidence in using AI has changed
- AI use in 2025 - interactive search engine
- AI use in 2026 - interactive search engine PLUS more fundamental tasks

Using AI as a code reviewer

I will post code files here. Please check for obvious errors. Do not re-write, give a bullet list of suggestions.

Check for obvious improvements

Check against the tidyverse style guide

- Good for spotting style inconsistencies and simple errors;
- no feel for an elegant/simple solution; no understanding of the context;
- good for following precise instructions; what exactly do you want it to do?

But how do you know what good code looks like?

What makes statistical code well-structured?

Give me ten tips for clean, well-organised statistical code

1. Organise code into clear sections
2. Use meaningful names
3. Keep each step simple
4. Comment the *why*, not the obvious
5. Keep important choices explicit
6. Run code in logical chunks
7. Check results as you go
8. Don't overwrite raw data
9. Avoid unnecessary repetition
10. Make the whole analysis reproducible

What makes R statistical code well-structured?

what about code specifically written in R tidyverse

1. Organise code into clear sections
2. Use meaningful `snake_case` names
3. Keep pipes short and readable
4. Break complex operations into logical steps
5. Comment the *why*, not the obvious
6. Keep important choices and parameters explicit
7. Handle missing values deliberately
8. Be explicit about grouping and ungrouping
9. Avoid repetition using functions and iteration
10. Check intermediate results as you go

So, how good is my code?

Here is some of my code. Check against the ten generic and ten tidyverse principles that are specified above

Yes. Looking across the R files you supplied, there is a surprisingly clear pattern: your code already embodies many of the principles very well, while a few recurring features give you excellent material for an AI code-review demonstration.

● Strengths

- Extensive checking throughout the analysis
- Clear structure and explicit parameters
- Good use of functions and iteration
- Raw and processed data kept separate
- Reproducible project structure

● Room for improvement

- Some generic object names: `temp`, `alpha`, `beta`, `gamma`
- Some long pipes and multi-step transformations
- Repeated analysis patterns that could potentially be parameterised
- Mixture of tidyverse and base R approaches

This is feedback from an LLM, not an expert opinion

Moving from generic feedback to specific suggestions

Using my files, give me one example of good coding and one example of bad coding

```
temp <- df %>%  
  group_by(app_no) %>%  
  summarise(n = n())  
  
id_temp <- temp %>%  
  filter(n > 1) %>%  
  select(app_no) %>%  
  pull()
```

Moving from generic feedback to specific suggestions

For each of the tidyverse principles generate one good example and one bad example. Examples need to be self-contained. Generate a dataset first if needed

```
students <- tibble(
  student_id = 1:5,
  gender = c("Female", "Male", "Female", "Male", "Female"),
  score = c(72, 61, 85, 68, 79)
)

# Prepare data
students <- students %>%
  filter(score >= 60)

# Summarise results
results <- students %>%
  group_by(gender) %>%
  summarise(
    n = n(),
    mean_score = mean(score),
    .groups = "drop"
  )
```

```
students <- tibble(student_id = 1:5, gender = c("Female", "Male", "Female", "Male", "Female"), score = c(72, 61, 85, 68, 79))
students <- students %>% filter(score >= 60)
results <- students %>% group_by(gender) %>% summarise(n = n(), mean_score = mean(score))
results
```

Make examples related to a (possibly hypothetical) cardiovascular trial

Using AI to generate teaching examples

I am writing a textbook on R for health economics. My main working example will be a synthetic RCT in Parkinson's disease. I need all my raw data to be in SDTM format. Assume it will have 50 patients in the treatment group and 50 patients in the placebo group. Generate several input datasets: demographics, randomisation split, follow-up visits, adverse events. include also the R code used for generation

```
# Synthetic Parkinson's disease RCT: SDTM-style raw data generator
# For textbook teaching only. All subjects, dates and events are fictional.

set.seed(20260917)

studyid <- "PD-HE-001"
ntreat <- 50
nplacebo <- 50
n <- ntreat + nplacebo

# -----
# Subject-level randomisation / demographics (DM)
# -----
subj <- sprintf("%03d", seq_len(n))
siteid <- sample(c("101", "102", "103", "104"), n, replace = TRUE)
subjid <- sprintf("%s-%s-%03d", studyid, siteid, seq_len(n))
```

Using AI to generate teaching examples

 AE	Microsoft Excel Comma S...	3 KB	No
 DM	Microsoft Excel Comma S...	2 KB	No
 generate_parkinson_rct_sdtm	R File	3 KB	No
 README	Text Document	1 KB	No
 SV	Microsoft Excel Comma S...	5 KB	No
 TA	Microsoft Excel Comma S...	1 KB	No
 TV	Microsoft Excel Comma S...	1 KB	No

1	2	3	4	5	6	7	8	9	10	11	12
STUDYID	DOMAIN	USUBJID	SUBJID	RFSTDTC	SITEID	AGE	AGEU	SEX	RACE	ETHNIC	COUNTRY
PD-HE-001	DM	PD-HE-001	1001	13/02/2026	102	62	YEARS	F	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1002	25/02/2026	104	57	YEARS	F	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1003	07/01/2026	101	79	YEARS	M	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1004	04/02/2026	103	63	YEARS	F	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1005	05/01/2026	102	68	YEARS	M	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1006	16/01/2026	101	72	YEARS	M	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1007	08/02/2026	101	65	YEARS	F	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1008	01/02/2026	101	71	YEARS	M	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1009	12/01/2026	102	72	YEARS	F	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1010	17/01/2026	104	68	YEARS	M	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1011	16/02/2026	102	70	YEARS	M	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1012	17/02/2026	101	71	YEARS	M	WHITE	NOT HISP/	GB
PD-HE-001	DM	PD-HE-001	1013	23/01/2026	104	70	YEARS	F	WHITE	NOT HISP/	GB

Using AI to adapt teaching examples

Rewrite this problem in the context of chronic kidney disease incorporating disease progression specifics. What would be the main differences?

- **Longer follow-up:** 3 years rather than 1 year to capture meaningful disease progression
- **Biomarker-driven progression:** repeated eGFR and UACR measurements replace neurological symptom scores as the main longitudinal measures
- **Patient-specific trajectories:** kidney function declines at different rates across patients, with treatment slowing progression
- **Clear disease states:** eGFR can be used to derive transitions between CKD stages and kidney failure
- **More interconnected data:** adverse events and clinical events depend on evolving kidney function rather than being generated independently
- **Additional SDTM domains:** laboratory (**LB**) data become central, alongside demographics (**DM**), visits (**SV**), and adverse events (**AE**)
- **Stronger link to economic modelling:** longitudinal trial data can naturally be transformed into health states for subsequent cost-effectiveness analysis

Using AI to learn about new concepts

explain in simple terms what intersectionality is

So in terms of statistical modelling what can I do? say I have a dataset of student exam results, plus information on student's gender, social class and ethnicity. How can I test for intersectionality? I use R

so how is intersectionality different from interaction?

what is MAIHDA framework?

this is pvc: 0.7865045. how should it be interpreted?

- All explanations come with R code;
- your questions can be as basic and repetitive as you'd like;
- **Check your answers and beware of long conversations!**

Code packaging with roxygen2

roxygen2 helps generate documentation / create packages.

Add roxygen documentation for R function <...>

translate function <...> into an R package

how to document dataset in r package

roxygenise <...>

- January 2023!
- Excellent experience, hours of mundane work saved;
- examples had to be fine-tuned to be made relevant;
- package was uploaded onto CRAN;
- same approach could be used to provide within-script commentary.

A sneak peek into VS code

VS code test

SOURCE CONTROL

The folder currently open doesn't have a Git repository. You can initialize a repository which will enable source control features powered by Git.

[Initialize Repository](#)

To learn more about how to use Git and source control in VS Code [read our docs](#).

You can directly publish this folder to a GitHub repository. Once published, you'll have access to source control features powered by Git and GitHub.

[Publish to GitHub](#)

generate_random_numbers.R

```

1 # Generate 1000 random numbers
2 # Author: Generated by GitHub Copilot
3 # Date: August 28, 2025
4
5 # Set seed for reproducibility (optional)
6 set.seed(123)
7
8 # Generate 1000 random numbers from uniform distribution
9 random_numbers <- runif(1000)
10
11 # Display first few numbers
12 cat("First 10 random numbers:\n")
13 print(head(random_numbers, 10))
14
15 # Save to CSV file
16 write.csv(random_numbers, "random_numbers.csv")
17
18 # Save to text file (one number per line)
19 writeLines(as.character(random_numbers), "random_numbers.txt")
20
21 # Display summary statistics
22 cat("\nSummary statistics:\n")
23 print(summary(random_numbers))
24
25 cat("\nFiles created:\n")

```

CHAT

I'll update the script to generate 1000 random numbers instead of 100.

generate_random_numbers.R +3 -3

Perfect! I've updated the script to generate 1000 random numbers instead of 100. The changes made:

1. Updated the comment to say "Generate 1000 random numbers"
2. Changed `runif(100)` to `runif(1000)`

The rest of the script remains the same - it will still show the first 10 numbers, generate summary statistics, and save the 1000 random numbers to both CSV and text files.

1 file changed [Keep](#) [Undo](#)

generate_random_numbers.R

[Add Context...](#) generate_random_numbers.R +

Add context (#), extensions (@), commands (/)

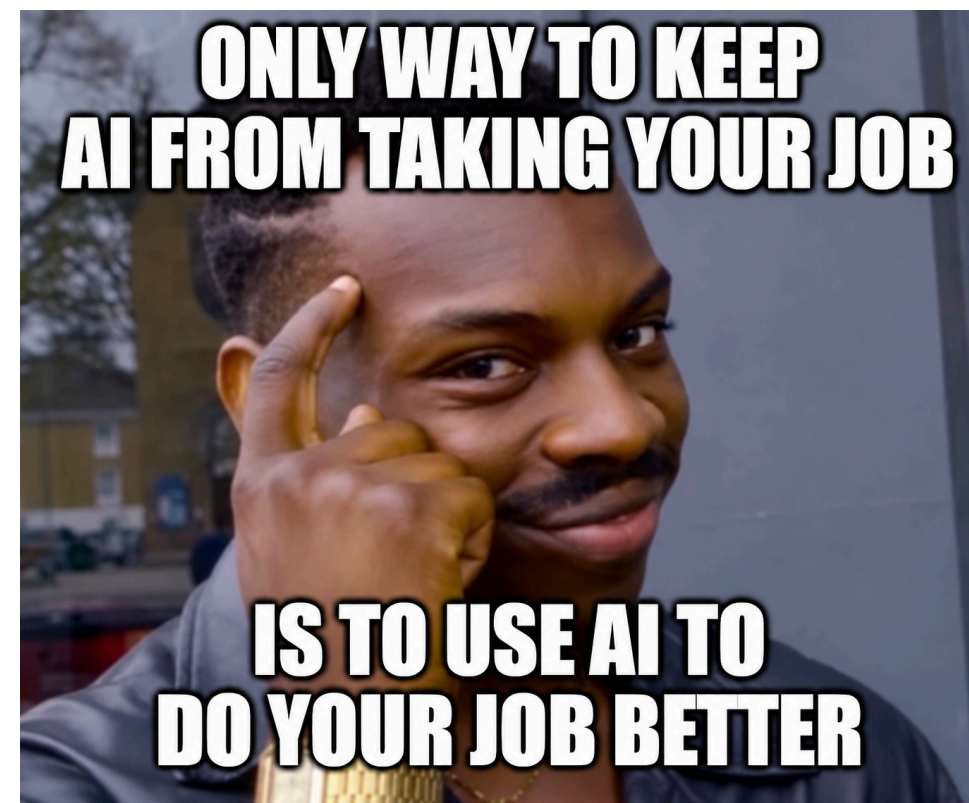
Agent Claude Sonnet 4

Keep Undo 1 of 1

What have I learnt?

What am I using AI for?

- Repetitive, (mundane), well-defined easy-to-check operations;
- coding help;
- code review against a set of rules;
- generating sets of rules for code reviews;
- learning new concepts;
- generating datasets, heavily commented code, teaching examples...



Source: <https://medium.com/@pondarajagadeesh98/the-ai-takeover-a-hilarious-look-at-the-future-54aad106af0a>

What am I NOT using AI for?

- Understanding data, setting up research framework and making decisions;
- writing code / text / presentations from scratch;
- tasks that are hard or impossible to check;
- loading identifiable-level data / confidential information;
- vibe coding (not yet anyway...);
- slide illustrations (though Hadley Wickham [disagrees](#))

Create a funny meme about chatGPT not having a sense of humour



Summary

- Incredibly powerful tool; generally positive experience; saves time and increases quality (and satisfaction!)
- A non-judgmental tutor available 24/7; polite, (overly) positive and patient. Motivating for learning new things
- The more effort goes into the prompt, the more useful the output. What is it you need?
- Vague requests could be good for brainstorming / organising thoughts
- If the answer is not helpful, is the issue AI or the prompt?
- **You need to read, understand and check the answers!**
- AI is a tool; but it is the human who poses the question and understands the context

Thank you!



GIF from <https://media.giphy.com/media/13HgwGsXF0aiGV/giphy.gif>

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